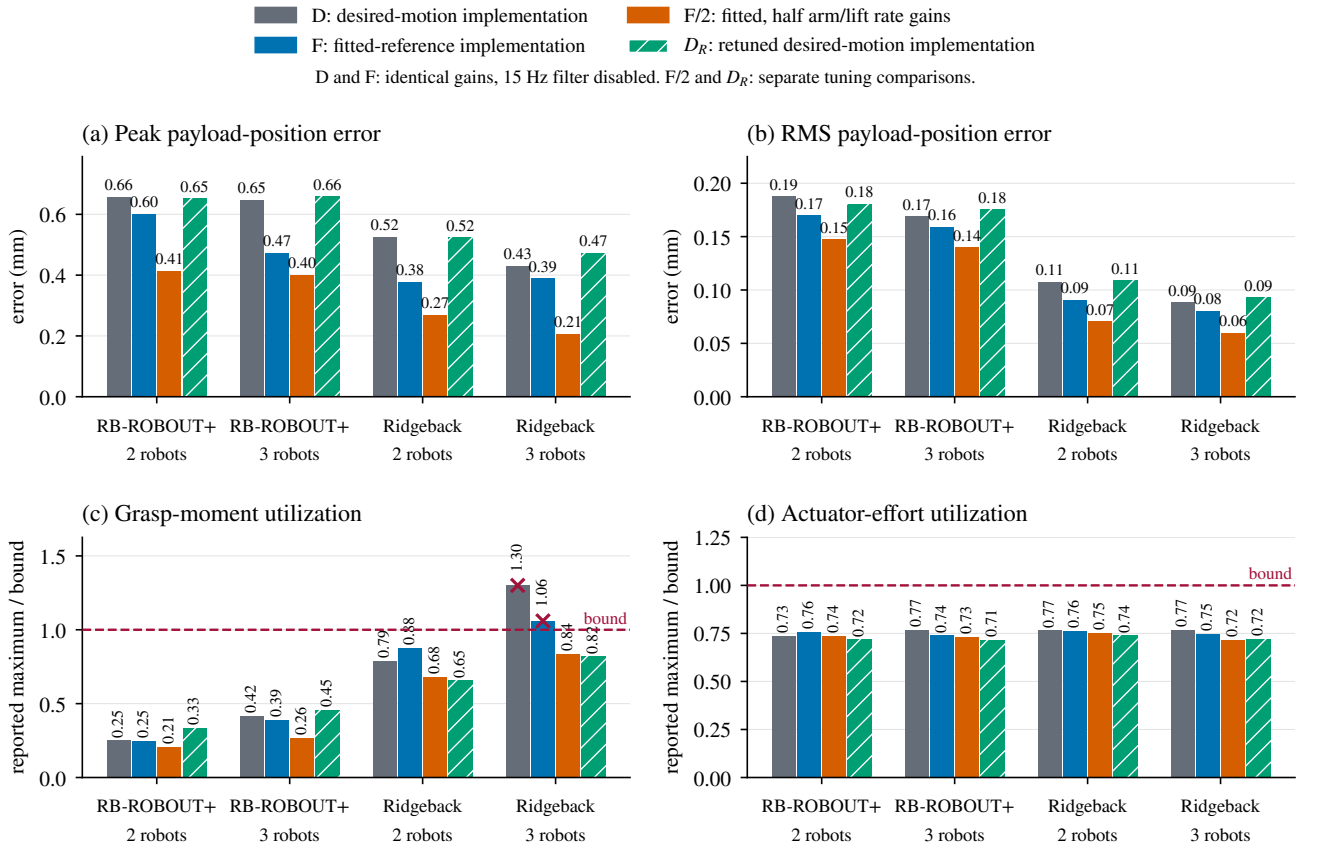
D: desired-motion implementation
F: fitted-reference implementation
F/2: fitted, half arm/lift rate gains
D_R: retuned desired-motion implementation
D and F: identical gains, 15 Hz filter disabled. F/2 and D_R: separate tuning comparisons.
(a) Peak payload-position error
error (mm)
0.66 0.60 0.41 0.65
0.65 0.47 0.40 0.66
0.52 0.38 0.27 0.52
0.43 0.39 0.21 0.47
(b) RMS payload-position error
error (mm)
0.19 0.17 0.15 0.18
0.17 0.16 0.14 0.18
0.11 0.09 0.07 0.11
0.09 0.08 0.06 0.09
(c) Grasp-moment utilization
reported maximum / bound
bound
0.25 0.25 0.21 0.33
0.42 0.39 0.26 0.45
0.79 0.88 0.68 0.65
1.30 1.06 0.84 0.82
(d) Actuator-effort utilization
reported maximum / bound
bound
0.73 0.76 0.74 0.72
0.77 0.74 0.73 0.71
0.77 0.76 0.75 0.74
0.77 0.75 0.72 0.72
RB-ROBOUT+ 2 robots
RB-ROBOUT+ 3 robots
Ridgeback 2 robots
Ridgeback 3 robots